Class: S

↳ class P

class C1 | class C2 | class C3

class P1 | class P2 — cannot extend more than 1 class

class C1

Polymorphism

is many ↳ forms ⇒ { method behavior changes based on method call } — Implementation

int name (int x, string y) { → body, behavior, ↳ implementation

return 5

}

should match

→ function / method

Compile time polymorphism

Runtime polymorphism [more imp]

① decision on which fn to call is made during compile time

② on what basis is the decision → name + parameters signature

Runtime Poly :- (Inheritance)

Class Parent {

func() {

≡

}

}

class child → Parent {

@override

func() {

≡ body 2

}

}

→ new implementation

overriding → changing the implementation of a inherited function by re-writing it in the child class

Runtime poly :- the decision to call a function P → C overriding is happening ↳ [on the type of object] → objects fn is called

P → C

P P1 = new C()

P1.func();

Dog

↳ do bark

C1 → doBark

C2 → dB

C3 → doB

C4 → doB

S → doB

Class DogSound {

sound (Dog d1) {

d1.doBark

}

}

sound (new C1) — Labrad

C3 → doBark